Spend Typo Budget Where the Classifier Is Unsure: Class-Balanced Margin-Targeted Augmentation

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Abstract

Typo augmentation can improve intent robustness, but uniform sampling spends a limited refit budget without asking which corruptions have low true-label score gaps. We propose Margin-Targeted Typo Augmentation (MTA), which generates four character-level variants per utterance, scores each under a clean classifier, and selects the lowest-margin candidates under equal per-intent quotas. On pinned CLINC150 subsets, we compare MTA with clean-only training, class-balanced random selection, and a four-times-larger balanced mixture. Across six prespecified settings evaluated with the same eight seeds held out from exploration, every paired bootstrap 95% interval for MTA over random selection is above zero. In the primary 50-intent setting, MTA improves mean noisy accuracy by 1.68 points (95% CI [1.32, 2.08]) and remains within one point of the full mixture while admitting 1,250 rather than 5,000 augmented examples to the final refit. The effect is positive for swap, deletion, insertion, and keyboard substitutions. These results establish a reliable empirical selector effect in this controlled setting, while leaving natural errors, neural encoders, unlabeled selection, candidate-generation cost, and causal mechanism open.

1 Introduction

Intent classifiers receive spelling variants that preserve intent but alter lexical evidence. Character edits can sharply reduce classifier accuracy (Pruthi et al., 2019), and mixed-noise training can recover part of this loss in goal-oriented dialogue (Sengupta et al., 2021). Which generated examples deserve a limited budget? Here, budget means records admitted to the final refit, not candidate generation and scoring. Figure 1 contrasts uniform selection with spending the same class-balanced refit count on the lowest true-label score gaps.

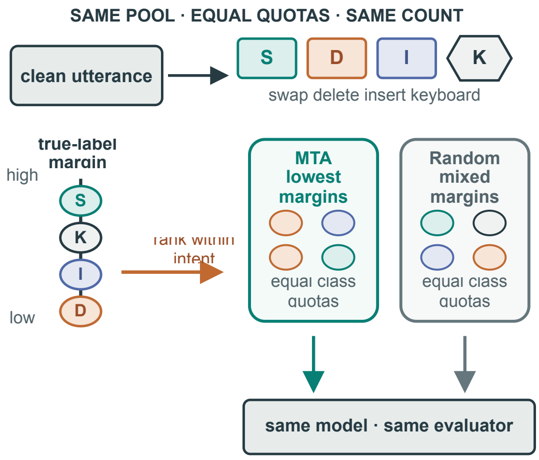

Figure 1: Random and margin-targeted selection use the same class-balanced refit count; MTA retains candidates with the lowest true-label score gaps.

Uniform sampling treats easy and label-confusing edits alike. We propose Margin-Targeted Typo Augmentation (MTA): generate one candidate per operator, score its true-label margin under a clean model, and retain the lowest-margin items under equal per-intent quotas. Only selection changes; the pool, model, example count, and evaluator stay fixed.

Across six prespecified settings sharing eight seeds held out from exploration, all paired bootstrap 95% intervals for MTA over random are positive. The primary 50-intent gain is 1.68 points [1.32, 2.08], and MTA stays within one point of an operator-balanced 5,000-example mixture while refitting on 1,250 examples. The claim is limited to CLINC150, a word-count classifier, English, and four synthetic edits. Section 3 defines MTA, Section 4 tests it, and Section 5 bounds mechanism and cost claims.

2 Related Work

Noisy intent understanding. Realistic input variation reduces both intent and slot performance, while aggregate augmentation improves average robustness (Sengupta et al., 2021). LAUG expands task-oriented dialogue stress tests to word perturbations, paraphrases, disfluencies, and speech phenomena (Liu et al., 2021). HINT3 and covariate-shift benchmarks further show that clean in-distribution scores do not establish deployment robustness (Arora et al., 2020; Khosla and Gangadharaiah, 2022). These studies motivate shifted evaluation; MTA asks how to allocate a fixed synthetic-noise budget.

Typo robustness and augmentation selection. Balanced insertion, deletion, substitution, and swap mixtures can transfer from synthetic to naturally occurring noise (Karpukhin et al., 2019), while character processing alone is not automatically robust (Belinkov and Bisk, 2018). Stable encodings, misspelling-aware embeddings, and compact projections attack the representation problem directly (Jones et al., 2020; Piktus et al., 2019; Sankar et al., 2021). Broader augmentation surveys and controlled utterance-classification studies document substantial variation among transformations (Chen et al., 2023; Xu et al., 2020). Decision-boundary sampling selects unlabeled text for annotation (Tong and Koller, 2001); uncertainty-guided generation has been studied for efficient active learning (Quteineh et al., 2020). MTA instead selects labeled synthetic typos under fixed quotas and refit counts. Word-form variation can alter lexical evidence (Wang et al., 2025), motivating model-specific rescoring.

3 Margin-Targeted Augmentation

3.1 Candidate margins

Let $D = \{(x_i, y_i)\}_{i=1}^{n}$ be the clean training set, O the four typo operators, and f_0 a classifier fitted on D. Each operator produces one deterministic candidate for a fixed seed,

$$c_{i,o} = T_o(x_i; s), \qquad o \in O. \tag{1}$$

Writing $g_k(c)$ for the pre-normalization score of intent k, we measure how securely the candidate retains its true intent by

$$m(c_{i,o}) = g_{y_i}(c_{i,o}) - \max_{k \neq y_i} g_k(c_{i,o}). \tag{2}$$

A negative margin means that the clean classifier assigns a competing intent a larger score. MTA uses low score gaps as a model-specific ranking signal, without consulting validation or test accuracy. The clean-only model computes the score before any augmented example is selected, so selection cannot benefit from a refitted classifier or final evaluation labels. Candidates remain separate records even when two operators emit the same surface form; this preserves the shared generation contract. The raw score gap is not a normalized geometric distance to a hyperplane. It can favor candidates with more edits or repeatedly select variants of one hard source, so the experiment tests the complete selector rather than isolating a boundary-proximity mechanism. Recomputing the ranking for a different representation is part of applying the method because score scales and competing intents may change. Ties follow immutable candidate order. Margin magnitude is only an ordering statistic: MTA assumes neither calibrated probabilities nor a tunable threshold.

3.2 Class-balanced selection

For total budget B and label set $\mathcal{Y}$, the quota for label y is

$$q_y = \left\lfloor \frac{B}{|\mathcal{Y}|} \right\rfloor + r_y, \qquad \sum_{y \in \mathcal{Y}} r_y = B \bmod |\mathcal{Y}|, \tag{3}$$

where the deterministic remainders $r_y \in \{0, 1\}$ follow label order. Within each label, MTA takes the q_y candidates with the smallest margin:

$$S_y = \mathop{\arg\min}_{S \subseteq C_y,\ |S| = q_y} \sum_{c \in S} m(c), \qquad S = \bigcup_y S_y. \tag{4}$$

This construction guarantees the same total count and intent distribution as the random-budget control. The class-balanced random baseline samples q_y candidates in each class; balanced full-mixture augmentation takes one example per clean item and distributes operators evenly. Every method refits the same multinomial Naive Bayes classifier. Without class constraints, global ranking could reweight difficult intents. All declared quotas are feasible because each class contributes four candidate streams per clean example. Quotas are fixed before ranking, so MTA and random differ in within-class membership, not class counts. This also makes class counts directly auditable. Selection adds one sort per class after shared candidate scoring and no additional model fit. The augmented-example ratio is computed against the full mixture.

3.3 Decision rules

We evaluate clean accuracy and accuracy for swap, deletion, insertion, and keyboard-neighbor substitution. Mean noisy accuracy averages the four operator accuracies within each seed; the worst-operator score takes their minimum. The primary comparison is the paired seed-level difference between MTA and random selection. Claim C1 requires the lower endpoint of its bootstrap 95% interval to exceed zero in every declared setting. Claim C2 uses a non-inferiority margin $\epsilon = 0.01$: if $\Delta_{\mathrm{MTA-full}}$ denotes paired mean noisy accuracy, support requires

$$\mathrm{LCB}_{.95}(\Delta_{\mathrm{MTA-full}}) > -\epsilon. \tag{5}$$

The bootstrap resamples the eight frozen perturbation seeds and recomputes the paired statistic; no setting or threshold is selected from final performance. Pairing is essential because all methods within a seed see perturbations derived from the same clean examples. The paired noisy-accuracy difference removes seed-level difficulty before resampling, whereas an unpaired interval discards this control. We retain unrounded per-seed accuracies and decide from percentile endpoints. Clean accuracy guards against robustness purchased by destroying uncorrupted performance, and the worst-operator score prevents a favorable mean from concealing one failed edit family. The primary scope takes the first 50 lexicographically ordered intents, giving 5,000 training, 1,000 validation, and 1,500 test utterances; a 20-intent scope tests label-space sensitivity. Additive smoothing is 1.0, and every selection policy receives the same candidate pool. This inferential design tests repeatability across perturbation randomness, not uncertainty from resampling the dataset or changing model families.

4 Experiments

4.1 Protocol

We use the official CLINC150 train, validation, and test partitions (Larson et al., 2019); intent labels make true-label margin and class quotas observable. The four typo operators are adjacent swap, deletion, insertion, and keyboard-neighbor substitution; this tests whether the selection gain spans distinct character edits. Each edit attempt has probability 0.45, with checks at 0.30 and 0.60. C1 covers 50- and 20-intent settings, both severity checks, and 10%/50% budgets at 20 intents. The full mixture contains one operator-balanced augmentation per clean item, not all four candidates. Seeds 307, 331, 353, 379, 401, 421, 443, and 467 are disjoint from exploration and fixed in the confirmation script. Each method's operator accuracies are persisted separately for paired reconstruction.

4.2 Primary comparison

Table 1 reports the method by metric comparison. At the matched 1,250-example budget, random selection reaches 0.8014 mean noisy accuracy, whereas MTA reaches 0.8182. The paired gain is 0.0168, with a 95% interval [0.0132, 0.0208]. MTA also raises clean accuracy from the clean-only model's 0.9133 to 0.9238. The full mixture remains 0.0036 higher on mean noisy accuracy, but the paired interval [-0.0056, -0.0017] lies entirely above the predeclared -0.01 boundary. Thus MTA meets the narrow refit-data claim without claiming superiority to full-mixture training. Appendix A records derivations and the executable check boundary.

4.3 Held-out-seed confirmation

Figure 2 plots each paired mean and C1 bootstrap interval. The six MTA-minus-random estimates are 0.0168, 0.0131, 0.0074, 0.0164, 0.0068, and 0.0174; every interval lower bound is positive. The smallest gain is at severity 0.30, but no margin-distribution analysis identifies why. At severity 0.60, MTA-minus-full is -0.0071 [-0.0127, -0.0021]; the interval crosses -0.01, so primary non-inferiority does not extend to this stress setting.

4.4 Operator analysis

Table 2 resolves the primary result by evaluation operator. Against random selection, MTA gains 0.0213 on swap, 0.0222 on deletion, 0.0096 on insertion, and 0.0138 on keyboard substitution. Thus the evaluation benefit appears on every operator. Relative to full mixture, MTA is lower on swap/deletion, higher on insertion, and nearly tied on keyboard; this descriptive pattern does not identify a training-side cause.

4.5 Budget sensitivity

Figure 3 fixes 20 intents and varies only budget. MTA exceeds random selection at 10%, 25%, and 50%, with gains of 0.0068, 0.0131, and 0.0174. These predeclared points show breadth, not a tuned optimum.

5 Discussion

The evidence supports the complete low-score-gap selector, not a geometric boundary mechanism. MTA and random share quotas, pool, refit count,

Method	Clean accuracy ↑	Mean noisy accuracy ↑	Worst-operator accuracy ↑	Augmented examples ↓
Clean only	0.9133	0.7948	0.7889	0
Random budget	0.9090	0.8014	0.7918	1,250
MTA	**0.9238**	0.8182	**0.8017**	1,250
Full mixture	0.9131	**0.8217**	0.7958	5,000

Table 1: Primary CLINC150 50-intent method-by-metric comparison, averaged over eight seeds held out from exploration. MTA uses the same 1,250-example refit budget as random selection; the full mixture uses 5,000 examples. Best values are bold.

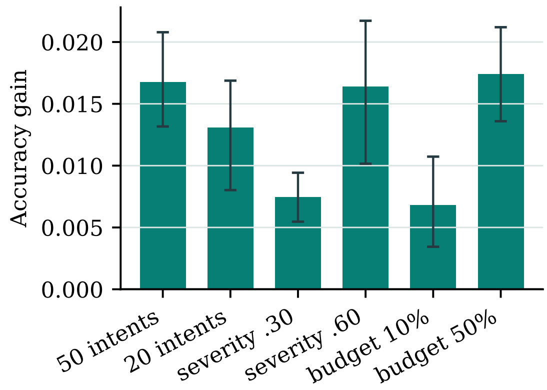

Figure 2: Confirmation setting by paired accuracy difference between MTA and the matched random budget. Error bars are bootstrap 95% intervals; every lower bound exceeds zero.

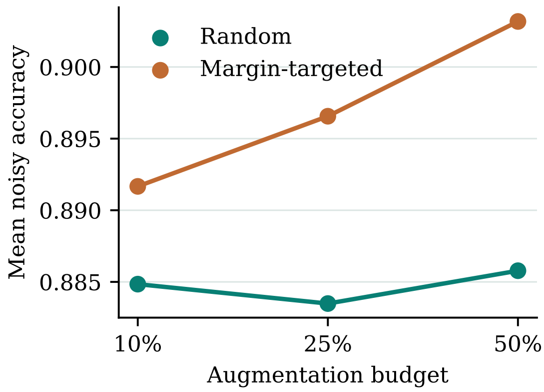

Figure 3: Budget by method and mean noisy accuracy in the fixed 20-intent scope. MTA exceeds matched random selection at every budget.

Method	Swap	Delete	Insert	Keyboard
Random budget	0.8114	0.8053	0.7922	0.7968
MTA	0.8327	0.8275	**0.8017**	0.8107
Full mixture	**0.8437**	**0.8363**	0.7958	**0.8113**

Table 2: Primary-setting method-by-operator accuracy. MTA improves over the matched random budget for every operator; best values are bold.

model, and tests, so their difference estimates a selection-policy effect. The policy may jointly alter edit severity and repeated-source weighting; this design does not separate them. Edit-matched selection, one-candidate-per-source selection, and clean hard-example duplication are the direct controls for future mechanism tests. Until then, breadth across operators and settings does not establish universal optimality or natural-error causality.

Refit-data efficiency is also not compute efficiency. The larger mixture is slightly stronger in primary noisy accuracy, while MTA meets the one-point rule and has the best primary clean and worst-operator means. MTA still generates and scores 20,000 candidates, so its advantage concerns records used for refitting. This may matter under storage or final-fit constraints, but the present CPU study does not measure wall-time or deployment savings. At severity 0.60, its interval against full mixture crosses the boundary; primary non-inferiority therefore does not generalize to every stress condition.

6 Conclusion

Margin-Targeted Typo Augmentation turns a fixed final-refit example budget into a supervised selection problem. On CLINC150, class-balanced low-score-gap selection improves noisy intent accuracy over matched random sampling in all six prespecified settings and remains within one point of a four-times-larger mixture in the primary setting. The result supports a narrow allocation principle for this classifier and candidate pool, while making no claim of lower candidate-generation cost or an isolated geometric boundary mechanism.

Limitations

This study uses English CLINC150 utterances, synthetic character edits, one word-count Naive Bayes classifier, and lexicographically selected intent subsets. True-label margins require labeled training data and cannot directly select unlabeled production errors. The operators approximate, but do not reproduce, naturally distributed keyboard, phonetic, morphological, or speech-recognition errors. The confirmation seeds quantify perturbation variability rather than dataset resampling or model-family uncertainty. Larger neural encoders may rank candidates differently, and the primary one-point non-inferiority result does not hold at every severity. These boundaries constrain the claim to a reproducible CPU study of augmentation allocation.

References

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A Reproducibility and Derivations

The CLINC150 file is pinned to revision `828f8093932c8fe6ca7936c3d2e52903b1c523de` and SHA-256 `36923c3705a59e08fe9c3883d8bc2dd966ef93e22cb78ac41171782a698d56e0`. Candidate generation, class quotas, score computation, selection, training, and evaluation are deterministic conditional on a seed. Each case uses 10,000 bootstrap resamples with persisted case-specific seeds from 20260817 through 20260822; comparison streams use deterministic offsets recorded by the implementation.

For a class with sorted margins $m_{(1)} \leq \cdots \leq m_{(|C_y|)}$, any size-$q_y$ subset omitting $m_{(j)}$ for $j \leq q_y$ while including $m_{(k)}$ for $k > q_y$ can reduce its sum by exchanging the two. Repeating the exchange yields the first q_y items, proving the class-wise minimizer in Eq. 4. Because the quotas sum to B and class candidate sets are disjoint, $|S| = B$. The executable verification checks recorded example counts and recomputes the two paired decision rules from persisted per-seed accuracies; deterministic rerunning is required to audit complete membership and record-level predictions.